

CURRICULUM VITAE

Arash Pashazadeh

Civil Engineer · Ph.D. Researcher in Systems
Engineering

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PERSONAL INFORMATION

Date of birth: 16 March 1990

EDUCATION AND TRAINING

2025–Present **Ph.D. Student, Systems Engineering**

Department of Civil, Coastal, and Environmental
Engineering, University of South Alabama, Mobile,
Alabama, USA.

Supervisor: Dr. Shenghua Wu.

2015–2018 **M.Sc., Civil Engineering — Water Engineering and Hydraulic Structures**

Department of Civil Engineering, Razi University,
Kermanshah, Iran.

Thesis: *Development of Hydrologic Models for Flow
Routing in Multiple-Branch Systems.*

Thesis committee evaluation: Excellent (19.25/20).

Supervisor: Dr. Mitra Javan.

2010–2014 **B.Sc., Civil Engineering**

Department of Civil Engineering, Kurdistan University,
Kurdistan, Iran.

2018 **HSE-MS Certificate**

Certificate No. 137IR0519 · 40 hours of training.

PUBLICATIONS

1. **Pashazadeh, A., & Javan, M. (2019).** Comparison of the Gene Expression Programming, Artificial Neural Network (ANN) and Equivalent Muskingum Inflow models in the Flood Routing of Multiple-Branched Rivers. *Theoretical and Applied Climatology*. Q2. DOI: 10.1007/s00704-019-03032-2.
2. **Pashazadeh, A., & Javan, M. (2019).** Improvement of lumped models in multiple inflows rivers by principal component analysis and reliability analysis. *Acta Geophysica*. Q2. DOI: 10.1007/s11600-019-00290-6.
3. **Pashazadeh, A., Javan, M., & Wu, S. (2025).** Development of Nonlinear Muskingum Models in Multiple Inflows River Systems. DOI: 10.21203/rs.3.rs-703757/v1.
4. **Pashazadeh, A., & Wu, S. (2025).** Issues in the Effect of Applied Explicit Numerical Solutions on the Accuracy of Nonlinear Muskingum Models. Ready to submit.
5. **Song, J., Pashazadeh, A., & Wu, S. (2025).** Advances in Microplastic Detection and Interception: A State-of-the-Art Review. Under review.

**DETAILED
PROFESSIONAL
DESIGN
RECORD**

2024

**Designer & Technical Office Engineer — TGEM
Construction & Hydropower Company**

Dushanbe, Tajikistan · Full-time

Sebzor Hydropower Project, Shokhdara River

- Designed monorail hoist systems on multiple intake shafts to facilitate equipment handling and maintenance.
- Designed concrete ferro-beam structures for the powerhouse to improve structural stability and load distribution under seismic conditions.
- Designed the MIV steel system of the powerhouse for safe and efficient water intake.
- Designed steel support holders for HVAC system installation inside the powerhouse.
- Designed thrust rings around the tailrace pipes and thrust-ring manifold to strengthen high-pressure flow zones.
- Completed structural design of concrete Shafts #1, #2, and #3 for the drainage system of the weir, with shaft depths of about 23 m.
- Designed the truss roof structure and crane-girder / rail systems of the powerhouse.
- Updated the fishpass-intake design to support safe fish migration while managing intake flows.
- Developed gabion-wall protection design on the right side of the weir to reduce erosion and protect structural integrity.
- Designed manhole concrete structures for three penstock sections to provide safe inspection and maintenance access.
- Designed steel holder structures for two permanent bridge ends.
- Prepared sample designs and material specifications for metal works to standardize fabrication and installation procedures.

2018–2024

Design Engineer — Abvarzan Consulting Engineers

Tehran, Iran · Part-time 2018–2020; Full-time 2020–2024

- **2022 · Ahvaz:** hydraulic modeling of pump stations.
- **2022 · Ahvaz:** hydraulic and structural design of a suction pond for a pump station (approximately 1,400 m²).
- **2022 · Ghadir Project:** design of stop logs and sluice gates of intakes.
- **2022 · Ahvaz:** design of an industrial steel warehouse structure (approximately 300 m²).
- **2022 · Ahvaz:** design of an industrial steel structure for a pump station (approximately 2,500 m²).
- **2021 · Ahvaz:** design of a concrete pump-station structure (approximately 2,500 m²).
- **2021 · Ghadir Project:** design of the concrete structure of a control room.
- **2021 · Ahvaz:** 3D modeling of approximately 15,000 m² of landscape and pump-station buildings using SketchUp.
- **2021 · Balarood Project:** design of stop logs and sluice gates of intakes.
- **2021 · Balarood Project:** design of an industrial steel structure for a pump station (approximately 2,500 m²).
- **2020 · Andimeshk / Balarood:** design of a diversion dam including spillway, ogee section, sediment pond, and related hydraulic components.
- **2020 · Andimeshk / Balarood:** design of an industrial steel structure of a pump station (approximately 300 m²).
- **2020 · Andimeshk / Balarood:** hydraulic and structural design of a suction-pond concrete structure for a pump station (approximately 250 m²).
- **2020 · Andimeshk / Balarood:** design of an industrial steel warehouse structure (approximately 250 m²).
- **2020 · Andimeshk / Balarood:** design of the concrete control room of the Balarood project.
- **2020 · Tehran:** design of three buried concrete water tanks (360, 216, and 360 m³) and a PRK elevated tank with concrete legs and a height of approximately 20 m.

- **2018 · Tehran:** design of the Kordan Dam control room and FRP reinforcement design.

2014–2020

Design Engineer — Arman Pei Alborz Consulting Engineers

Tehran, Iran · Part-time

- **2018 · Tehran:** design of the steel structure of a sports club (approximately 2,000 m²).
- **2017 · Pakdasht:** design of water-supply and sewage pipe networks for the environmental-relocation project transferring pollution-related jobs from Tehran to Pakdasht.
- **2017 · Pakdasht:** design of the street network for the same environmental-relocation project.
- **2016 · Tehran:** design of water-supply and sewage pipe networks for an approximately 90,000 m² urban-planning project in the 6th phase of Parand.
- **2016 · Noorabad-Delfan:** design of water-supply and sewage pipe networks for an approximately 80,000 m² urban-planning project.
- **2015 · Zanzan:** design of a 16-floor residential reinforced-concrete building.
- **2014 · Noorabad-Delfan:** member of the design team for 1,200 residential units in an approximately 80,000 m² urban-planning project.
- **Various years:** design of reinforced-concrete retaining walls.
- **Various years:** design of pipeline valves and their concrete structural chambers.

Additional scope **Related engineering work reflected in earlier CV versions**

- Design of concrete and steel structures of agricultural pump stations.
- Design of agricultural water and drainage networks.
- Design of hydraulic structures for agricultural equipment, including valve chambers and small check dams in farm canals.
- Designer and technical-office engineering in steel/ concrete structures, hydropower, water, and civil infrastructure.

RESEARCH AND WORK INTERESTS

Numerical hydrologic and hydraulic modeling; flood forecasting; stormwater management; porous media; machine learning; artificial intelligence programming; aquatic and non-aquatic structures design; coastal and floating systems; CFD; sensing; experimental hydrodynamics; professional programming for civil engineering applications; network design; and web development.

TECHNICAL AND DIGITAL SKILLS

Water / Hydraulics: Flow 3D, EPANET, HEC-RAS, HAMMER, WEAP, Water/ Sewer CAD; hydraulic and hydrologic flow-routing models; pump-station hydraulics; agricultural water and drainage systems; water-supply structures.

Programming / Data: Python, MATLAB; machine learning, optimization, signal processing, IMU analysis, and engineering-code development.

CFD / Research: OpenFOAM, numerical modeling, wave-flume testing, experimental hydrodynamics, sensing, and data synthesis.

Structural / Civil: SAP2000, SAFE, ETABS, ABAQUS, Civil 3D, PROKON; steel and reinforced-concrete structures, industrial structures, retaining walls, and support structures.

Geotechnical: GeoStudio / Geo Office tools including SEEP/W, SLOPE/W, SIGMA/W, QUAKE/W, and CTRAN/W; PLAXIS.

Artificial Intelligence / Intelligent Systems: ANN (MLP, RBF), GEP, GP, GeneXproTools, NeuroSolutions, ANFIS, and CANFIS.

Optimization / Statistics: Genetic Algorithm, Particle Swarm Optimization, Ant Colony Optimization, Bat Algorithm; factor analysis, risk analysis, regression analysis, reliability analysis, and principal component analysis.

Drawing / Visualization: AutoCAD, SketchUp, Photoshop.

General: Microsoft Project, Excel, Word, PowerPoint, Publisher.

Web / Network / Civil Programming: professional programming for civil-engineering applications, network design, and web development.

LANGUAGE SKILLS

- **English:** TOEFL iBT 104/120 (16 Sep. 2024): Reading 26/30, Listening 28/30, Speaking 21/30, Writing 29/30.
- **Earlier TOEFL record:** 107/120 (26 Apr. 2022): Reading 30/30, Listening 30/30, Speaking 19/30, Writing 28/30.
- **Persian:** Fluent in oral, aural, and writing.
- **Kurdish:** Fluent in oral, aural, and writing.
- **Arabic:** Fluent in aural and writing.

HONORS AND MEMBERSHIPS

- Ranked 30th nationwide in Iran (2018) in the Ph.D. candidate selection examination for Water Resources Management Engineering, evaluated by the National Organization of Educational Testing (NOET).
- Member, Tehran Construction Engineering Organization (TCEO), Membership No. 1030142105.
- Member, Iran Value Engineering Association, Membership No. 381452.

WEBSITE NOTE

This About page is the consolidated CV-style version built from the uploaded CV files. The separate Research, Projects, Publications, Conferences, and Contact pages in the site remain available for ongoing updates through the Admin panel.

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